



RAMCO INSTITUTE OF TECHNOLOGY (An Autonomous Institution)

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Department of Information Technology **Academic Year 2025 - 2026 (Even Semester)**

Semester, Degree & Branch: VI Semester, B.Tech. IT

Course Code & Title: CCS354 & Network Security

Name of the Faculty member (s): Mrs. G.Sivasathiya, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit V / Types of Firewalls
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO14
- **Activity Chosen:** Brain Teasers
- **Justification:**

Brainteaser activities encourage students to think critically, analyze security concepts creatively, and improve problem-solving abilities. Conducting brain teaser activities on firewall concepts helps students understand different firewall types in an interactive and engaging manner. This activity promotes active participation, enhances logical reasoning skills, and strengthens students' understanding of network security concepts.

- **Time Allotted for the Activity:** 15 Minutes

- **Details of the Implementation:**

- The topic "Types of Firewalls" was selected for the brain teaser activity. Students were divided into small groups and provided with firewall-related brain teaser questions and scenario.
- The activity included puzzles, reverse-thinking questions, and scenario-based challenges related to packet filtering firewalls, proxy firewalls, stateful inspection firewalls, and next-generation firewalls.
- Students discussed among themselves and collaboratively solved the given challenges. The resource person of the workshop guided students during the activity and clarified concepts wherever necessary. Fig 1. shows students participating in the brainteaser activity on firewall concepts which was conducted by resource persons.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO5	2	1	3	2	2	2	1	1	3	2	1	1

(1 – Low

2 – Moderate

3 – High)

PO/PSO Mapped:

Innovative Practice	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO5	2	1	3	2	2	2	1	1	3	2	1	1
Justification for Correlation	Applying real time techniques with security tools requires a solid understanding of engineering principles to develop robust and efficient security solutions for protecting networks and systems	Utilizing security practices involves analyzing security requirements and identifying suitable automation strategies to address vulnerabilities and threats effectively	Applying security practices entails designing and developing application scripts and frameworks to validate network systems and protect against vulnerabilities comprehensively.	Real Time security tools involve investigating complex security threats and devising automated approaches to detect and mitigate vulnerabilities effectively	Leveraging real time application security tools demonstrate proficiency in utilizing modern security technologies to enhance the efficiency and effectiveness of threat detection and vulnerability management	Applying application security tools requires collaboration within teams and individual contributions to design, develop, and execute security scripts to safeguard systems.	Effective communication of application security strategies and results to stakeholders and team members ensures alignment and support for security automation initiatives	Applying security tools involves managing resources, timelines, and budgets effectively to ensure the successful implementation of real time application security measures and threat mitigation processes.	Applying security tools necessitates a commitment to continuous learning to stay updated with evolving security threats, techniques, and best practices.	Applying security practices for real time applications demonstrates proficiency in network security, ensuring comprehensive protection of systems and identifying and providing solutions for complex security challenges	Developing the network security with modern security practices aligns with the application of advanced technologies, contributing to the reliability and protection of IT systems	Applying security practices showcases proficiency in securing network systems using open-source technologies to deliver reliable and robust solutions for business and industry needs

- **Images / Screenshot of the practice:**



Figure 1. Students participating in the brainteaser activity conducted by resource persons.

Reflective Critique:

❖ ***Feedback of practice from students and other stakeholders:***

- Students found the activity interesting and enjoyable as it combined learning with problem-solving.
- Students felt that solving puzzles and scenario-based questions improved their understanding of firewall concepts..
- It was observed that student participation was improved, collaboration, and critical thinking during the activity.

❖ ***Benefit of the practice:***

- The activity improved logical reasoning and critical thinking skills.
- Students developed better understanding of different firewall types and their real-world applications.
- Collaborative learning during the activity enhanced teamwork and communication skills

❖ ***Challenges faced in implementation:***

- Some students initially required guidance to understand the problem-solving approach.
- Managing time for discussion and solution explanation was slightly challenging.
- A few students hesitated to participate actively during the initial stage of the activity.

❖ **References:**

- <https://www.cloudflare.com/learning/firewalls/what-is-a-firewall/>
- <https://www.fortinet.com/resources/cyberglossary/types-of-firewalls>
- https://www.cisco.com/c/en_in/products/security/firewalls/what-is-a-firewall.html

Signature of Faculty Member

HOD